

Software Design & Analysis (CS3004)

Date: April 8, 2026

Course Instructor(s)

AR, AI, ZN

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

solution

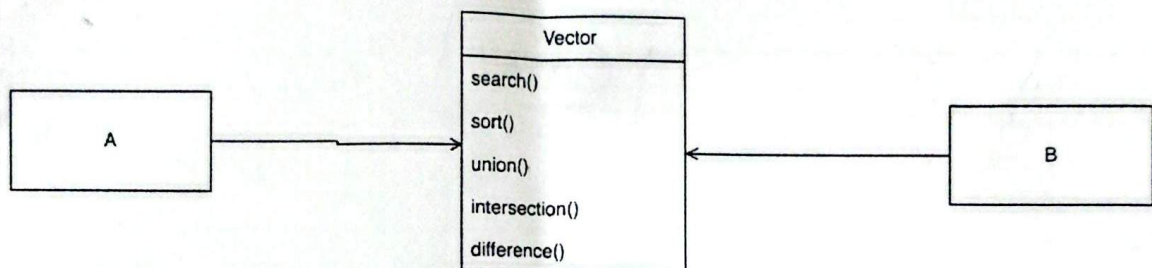
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Section

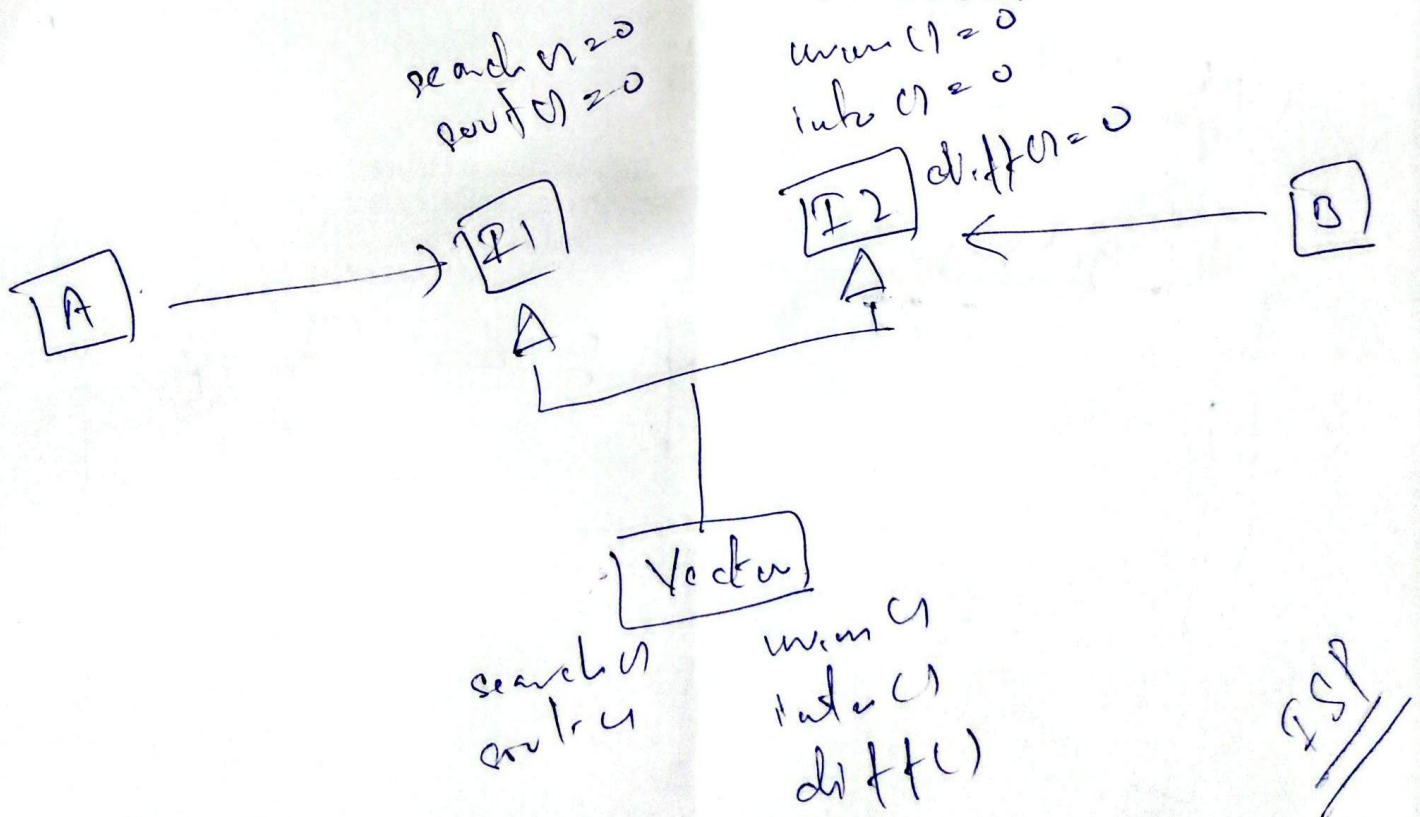
Student Signature

*Write your answers in the space provided. No separate answer sheet required for this exam!*CLO 5: Describe software design guidelines and principles

Q1: Consider the following class diagram:



Here, class A uses only search and sort, while class B calls only union, intersection and difference. Apply appropriate SOLID principle(s). Show your final design (class diagram).

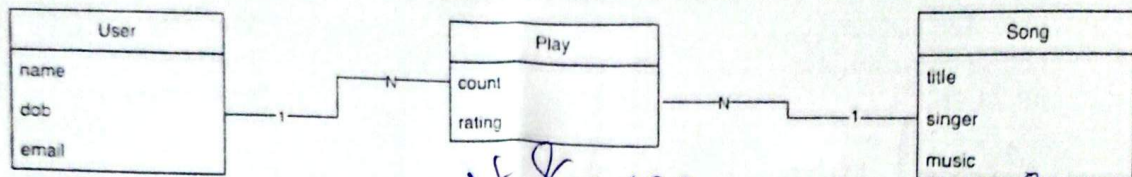


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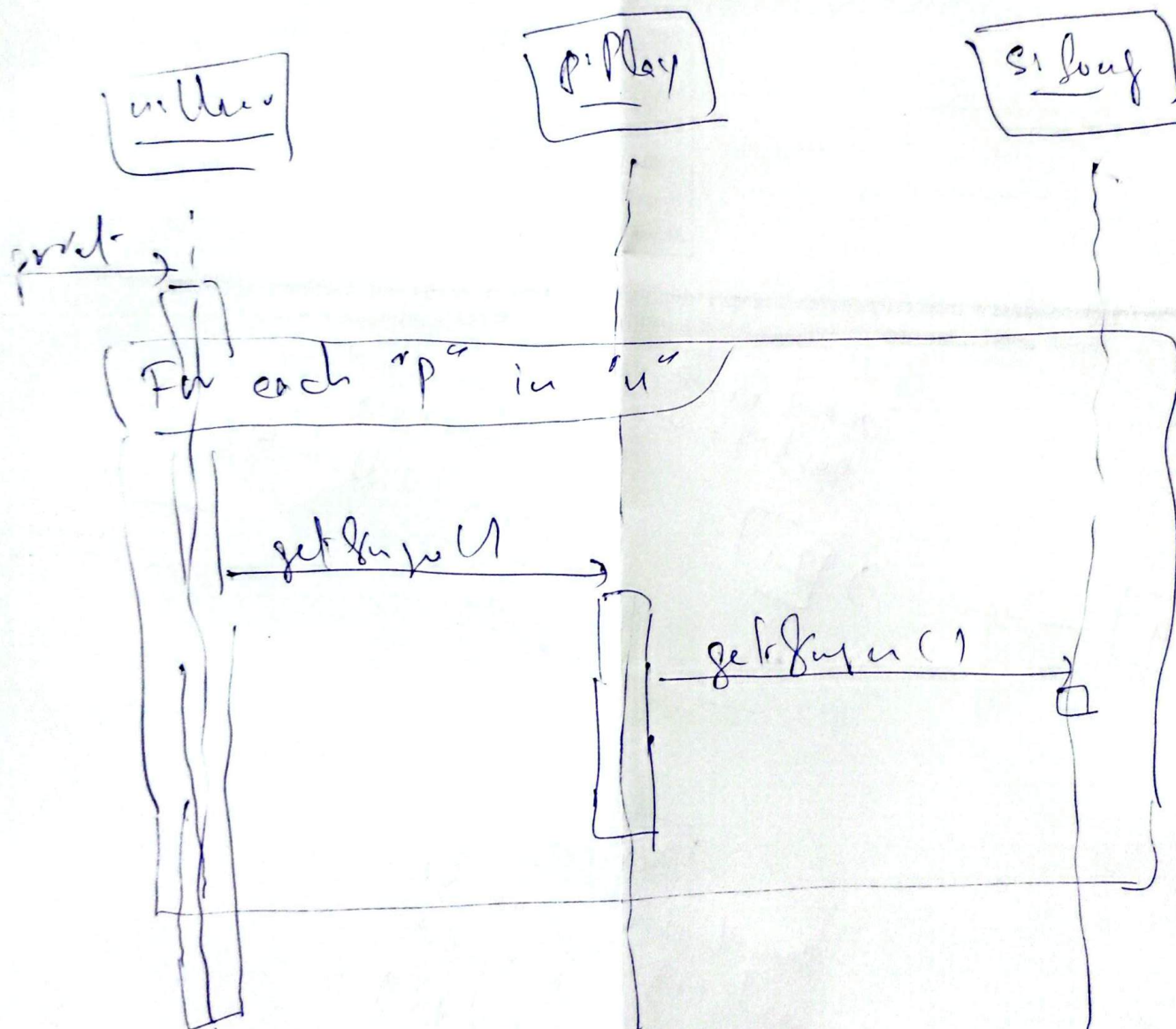
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CLO 3: Use different UML notations for software design

Q2: Consider the following class diagram:



Now give a sequence diagram to print all the singers for a given user (the singers whose songs were played by the user). You may add a few functions in the above classes. Assume the first function called is `User::print()`

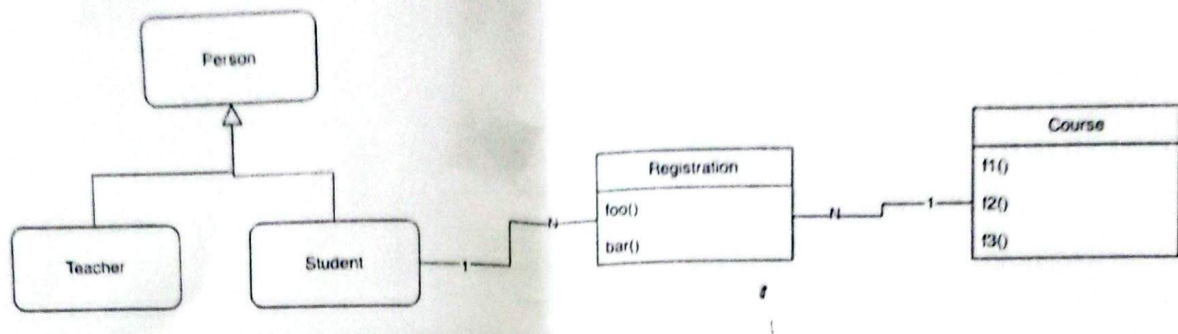


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CLO 2: Implement object-oriented principles for software analysis and design

Q3: Consider the following class diagram:



Now fill the following table:

	WMC	CBO	DIT	NOC
Person	0	0	0	2
Teacher	0	0	1	0
Student	0	1	1	0
Registration	2	2	0	0
Course	3	1	0	0

Here,

WMC = Weighted methods per class

CBO = Coupling between objects

DIT = Depth of inheritance tree

NOC = Number of children